

Jesus Andres Ruiz

jesus.ruiz4@my.utsa.edu | (956)-290-6269 | www.linkedin.com/in/jesus-andres-ruiz

Education

University of Texas at San Antonio

Bachelor of Science in Mechanical Engineering | GPA: 3.6

UTSA Honors College

San Antonio, TX

August 2024 – May 2028

Skills

Software & Tools: SolidWorks, Fusion 360, FEA, Microsoft Office

Programming / Embedded: Arduino, MATLAB, Python, Git/Github, ROS (learning)

Manufacturing & Design: DFM, GD&T, Design Iteration, Rapid Prototyping, Data Evaluation

Fabrication & Tools: FDM, Wiring/Soldering, Mig/Tig Welding, Mill, Lathe

Relevant Experience

Intelligent Systems and Bionics Engineering Lab

Undergraduate Research Assistant

San Antonio, TX

May 2026 – Present

- Led fabrication of the OpenExo hip module, an open-source wearable exoskeleton, working closely with graduate students to fabricate and assemble hardware across 3D printing and machining.
- Supported a graduate student's Bowden cable transmission project, contributing to prototype design and assembly for ankle actuation.
- Diagnosed and resolved 3D printer hardware/adhesion issues across 2 printer platforms (FDM and continuous carbon fiber), restoring consistent prototyping output.

UTSA Hypersonics Lab

Undergraduate Research Volunteer

San Antonio, TX

January 2026 – May 2026

- Fabricated precision diaphragms in the machine shop for use in a Mach 7.2 Ludwig Tube wind tunnel, supporting high-speed flow experiments and ensuring reliable burst pressures for test runs.
- Assisted researchers during hypersonic wind tunnel testing, helping prepare instrumentation, monitor runs, and collect experimental data from the wind tunnel.
- Set up and aligned laser-based Schlieren imaging systems to visualize shock waves and boundary-layer structures in hypersonic flow experiments.

UTSA Formula Society of Automotive Engineers

Mechanical Engineer / Team Member

San Antonio, TX

August 2025 – Present

- Designed and iterated mechanical components using SolidWorks, applying design for manufacturing principles and engineering drawings.
- Collaborated with a team of 5 to refine designs under competition constraints.
- Translated theoretical engineering concepts into real-world vehicle systems.

Project Experience

Open Exo Hip Module

Undergraduate Research Assistant – Project Lead

San Antonio, TX

May 2026 – Present

- Re-designed and iterated hip module components in Fusion 360, using FEA to validate redesigns after early prototype failures.
- Manufactured parts via 3D printing (FDM, continuous carbon fiber), CNC machining, and outsourced cutting; corrected a 200-micron shoulder screw bore tolerance issue caused by anodizing buildup before final assembly.
- Iterated the hip abductor through 3 design/print revisions before finalizing it as a machined, anodized 6061-T6 aluminum part, incorporating lessons from structural failure analysis on earlier cracked cuffs, sliders, and struts.

Bowden Cable Transmission – Ankle Actuation

Undergraduate Research Assistant

San Antonio, TX

May 2026 – Present

- Prototyped a Bowden cable transmission enabling remote ankle actuation, reducing distal leg mass and metabolic cost of walking.
- Designed, printed, and tested 2 TPU cable insert geometries (tapered vs. straight) to optimize routing and fit for cables.